VOXTURR Labs

PRODUCT DEFINITION · V0.1

School HRMS

A human-resource & payroll platform built for Indian educational institutions

Competitor analysis · functionality blueprint · foundational data model & key workflows for the first release.

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PREPARED BY	PREPARED FOR	DATE	CLASSIFICATION
Voxturr Labs	Internal	15 July 2026	Confidential

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This document is a working product-definition draft prepared to finalise scope ahead of the full technical specification. All figures for statutory rates, ceilings and pricing are indicative and are held in configurable form in the product.

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Document Control

Metadata

Title	School HRMS — Product Definition
Version	0.1 (Draft)
Date	15 July 2026
Prepared by	Voxturr Labs
Prepared for	Internal
Classification	Confidential
Status	Scope finalised (D1–D7); foundation & workflows drafted

Change log

Version	Date	Summary
0.1	15 Jul 2026	Competitor research (Keka, Zoho People, greytHR, HROne), scope finalisation, functionality blueprint, and foundational data model & workflows.

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Executive Summary

The opportunity

Every school and college in India runs on staff — teachers, administrators, support and transport personnel — yet the HR and payroll software they use is built for corporate offices, not classrooms. The market leaders (**Keka, Zoho People, greytHR, HROne**) are capable corporate platforms that schools *adapt to*, working around the mismatch between a corporate calendar and an academic one, between office staff and teaching staff, between a monthly salary and a per-lecture honorarium.

We are building an **HRMS purpose-built for Indian educational institutions** — starting with **private, multi-branch K-12 school groups** — that matches those incumbents on compliance-grade payroll while leading on the things none of them do.

What makes it different

Four capabilities, structural to schools, are absent from every competitor we studied:

1. **Academic-calendar-driven leave** — leave cycles follow the academic session, term and vacation structure, not the January–December corporate year.
2. **Vacation-staff vs. non-vacation-staff logic** — teachers receive long vacations *in lieu of* earned leave, while 12-month office staff accrue it normally. This single distinction changes both leave *and* salary calculations.
3. **Multi-scale payroll** — regular salaried staff and per-lecture / per-period guest faculty are paid in the same payroll run, each by its own logic, with statutory correctness throughout.
4. **Substitute-teacher arrangement** *(Phase 2)* — when a teacher is absent, the system arranges cover from the live timetable and feeds the substitution honorarium into payroll. No competitor offers this.

Items 1–3 ship in the first release; item 4 follows in Phase 2, and the data model is designed for it from day one.

MVP in one line

> A **compliance-grade Indian staff-payroll HRMS that is unmistakably built for schools** — matching greytHR/Keka on statutory payroll across multiple campuses, and leading with academic-calendar leave, vacation-staff logic, and multi-scale payroll.

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Scope at a glance

Dimension	Decision
Market	India — full statutory stack (PF, ESI, Professional Tax, TDS, gratuity, bonus, LWF)
First customer	Private / self-financed K-12 schools, **multi-branch** groups and trusts
Long-term scope	Any educational institution — pre-school, K-12, coaching, college/university (schema designed for it; MVP targets K-12)
Delivery	Multi-tenant SaaS (default) + single-tenant on-premise for enterprise clients
Clients	Web (browser) + mobile app for staff self-service
Full hire-to-retire in v1	**Recruitment/ATS is in the first release** — requisition → posting → shortlist → interview → offer → auto-convert to employee — so the joiner journey is complete end-to-end, not bolted on later
Governed employee records	HR uploads staff documents at enrollment; employees **cannot self-edit** — any change flows through an **HR-initiated request → employee upload → HR verification**, fully audited

Technology at a glance

Layer	Choice
Backend	Django + Django REST Framework, PostgreSQL
Web front-end	Next.js (React / TypeScript)
Mobile	Flutter
API	strawberry-django **GraphQL** (primary) + **DRF REST edge** (ingestion/files/auth/integrations); GraphQL SDL + OpenAPI generate the web (TypeScript) and mobile (Dart) clients, and are exposed to school-ERP integrators
Hosting	Multi-tenant SaaS; single-tenant on-premise from the same codebase
Team	~2 engineers working with AI agents — favouring proven, low-maintenance technology

Delivery approach

The build is **layered so each stage depends only on those before it** — tenancy and org structure first, then identity, people, the academic calendar, leave, attendance, payroll, statutory compliance, and finally self-service. This lets a small team deliver a coherent, testable product incrementally, and lets the business demonstrate value at each layer rather than waiting for a big-bang release.

The remainder of this document covers the competitor analysis that informed scope, the full functionality blueprint (every feature tagged MVP / Phase 2 / Later), and the foundational data model and workflows for the first release.

Competitor Analysis — HRMS Reference Products

Research basis for the school HRMS. Four India-first HR platforms were studied for feature depth, strengths, weaknesses, pricing, and fit for educational institutions. **Key takeaway: all four are horizontal corporate HRMS with *no* education edition** — they handle staff back-office HR well but model none of the academic-domain needs (substitution, academic-calendar leave, vacation-staff pay, period-level attendance, timetable/SIS linkage). That gap is our product thesis.

Side-by-side comparison

Dimension	Keka	Zoho People	greytHR	HROne
Vendor	Keka Technologies (Hyderabad)	Zoho Corp	Greytip Software	HROne (Uneecops)
Positioning	All-in-one, design-led HRMS	Suite-integrated, low-cost HRMS	India payroll & compliance specialist	All-in-one India HCM, mobile-first
Native payroll	✅ Strong, compliance-first	❌ Needs separate **Zoho Payroll**	✅ Deepest (30+ yrs)	✅ Strong, zero-touch
India statutory (PF/ESI/PT/TDS/LWF)	✅ ECR, 24Q, challans, signed F16	Via Zoho Payroll	✅ 24Q+FVU, signed F16/12BA, multi-state PT	✅ Both TDS regimes, ECR, challans
Native ATS	✅ Keka Hire (add-on)	❌ Needs **Zoho Recruit**	✅ greytHR Recruit (AI, Lisa bot)	✅ (Enterprise tier)
Attendance modes	8 modes, 200+ devices	Biometric, facial, geo/IP	Biometric, geo (Geotrack/fence), mobile	Geo-fence, selfie, biometric, **WhatsApp bot**
Performance/PMS	✅ Mature (OKR, 360, calibration)	✅ KRA/OKR/360	Add-on (~₹35/user)	✅ 9-box, succession
LMS	Keka Learn (add-on, staff)	✅ Enterprise tier (staff)	❌ (Academy = HR content only)	✅ Staff training
Mobile app	Solid (lags web)	Native iOS/Android	Mobile-first, Bella bot	**Award-winning, 110+ tasks**
Multi-entity / multi-branch	✅ Multi-legal-entity	Custom orgs	✅	✅ Multi-legal-entity
Pricing model	₹9,999–15,999/mo band (≤100 staff) + add-ons, setup fee, 18% GST	**1.25–4.50/user/mo** (5-user min); payroll/ATS separate	Per-user, India-focused, tiered	Base ₹4,950/mo (50 users) + ₹99+/user; 50-user min

Dimension	Keka	Zoho People	greytHR	HROne
Best-fit segment	Mid-market, adoption-sensitive	SMB / already-in-Zoho, budget	Compliance-heavy India payroll	50–150 employees
Review avg	~4.5 G2	Value-leader, mixed UX	Strong on compliance	~4.2–4.4 aggregate

Per-product notes

Keka

- **Strengths:** Cleanest UI (low training overhead), true hire-to-retire integration, compliance-first payroll, very flexible attendance (8 modes / 200+ devices), mature performance module.
- **Weaknesses:** Limited customization for non-standard processes; support/turnaround complaints; mobile lags web; many high-value modules are paid add-ons; effective cost ~35–55% above headline (setup fee 2× monthly, 18% GST, quarterly minimum billing).
- **Education fit:** Has a dedicated Education industry page — fits staff shift-scheduling, leave→payroll, biometric campus attendance. No SIS/academic modules; Keka Learn is corporate training, not student LMS.

Zoho People

- **Strengths:** Lowest cost/user, broad module coverage, deep Zoho-suite integration, highly customizable (no-code forms/workflows/custom apps), Zia AI, generous free tier (≤5 users).
- **Weaknesses:** Steep setup/learning curve ("feels like you need a developer"), cluttered DB-heavy UI, **no native payroll or ATS** (needs Zoho Payroll + Zoho Recruit), LMS & Helpdesk gated to Enterprise, weaker non-Zoho integrations.
- **Education fit:** Generic HR fit only; good for budget-conscious schools already in Zoho. Must pair with a school ERP/SIS. India statutory via Zoho Payroll.

greytHR

- **Strengths: Deepest India payroll & statutory compliance** (Form 24Q + FVU validation, signed Form 16/12BA, multi-state PT, LWF, ECR, PayNow disbursement), 150+ admin/MIS registers, KYE staff dossier, mail-merge letters, strong mobile-first ESS.
- **Weaknesses:** PMS is a newer add-on (less mature); no embedded corporate LMS; doesn't sell biometric hardware (integrates only).
- **Education fit:** Compliance backbone maps well to schools/trusts; leave engine flexible (comp-off, maternity, sandwich/restricted holidays, leave-year-end). No academic modules.

HROne

- **Strengths:** Deep India statutory out-of-box, 127 HR processes in one suite, **award-winning mobile app (110+ tasks, WhatsApp bot)**, budget-friendly transparent pricing, robust integrations (Tally/SAP/Oracle, 200+ biometric families, REST/webhooks), One AI voice agents.
- **Weaknesses:** Aggregate ratings more moderate than vendor claims; UI learning curve; mobile performance complaints; rigid/limited custom audit reports; advanced modules only in Enterprise tier; 50-

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user minimum.

- **Education fit:** Generic back-office fit (attendance, payroll, leave, ESS, multi-entity for trusts). No substitute-teacher, credential tracking, tenure, or timetable/academic-calendar linkage. Must pair with a school ERP/SIS.

Per-module capability matrix

Legend: ● native & strong · ◐ present / partial · ○ add-on or separate product · ✕ absent

Module	Keka	Zoho People	greytHR	HROne	Our target (MVP)
Core HR / HRIS	●	●	●	●	● + education staff taxonomy
Multi-entity / multi-branch	●	◐	●	●	● group→trust→campus
Attendance (biometric/geo/selfie)	●	●	●	●	● (+ half-day)
Period-level attendance	✕	✕	✕	✕	Phase 2
Leave management	●	●	●	●	● academic-calendar
Vacation-staff leave logic	✕	✕	✕	✕	●
Payroll engine	●	○ (Zoho Payroll)	●	●	●
Per-lecture honorarium payroll	✕	✕	✕	✕	●
India statutory (PF/ESI/PT/TDS)	●	◐	●	●	● full parity
Signed Form 16 / 24Q / ECR	●	◐	●	●	●
Gratuity / bonus / LWF	◐	◐	●	●	●
Full & Final settlement	◐	◐	●	●	●
Substitute-teacher arrangement	✕	✕	✕	✕	Phase 2 (schema-ready)
Performance / appraisal	●	●	○	●	Phase 2 (education-shaped)
Recruitment / ATS	○	○	●	◐	**MVP** (requisition→offer→convert)
HR-controlled document governance	◐	◐	●	◐	● MVP (request-gated employee updates)
LMS / training	○	◐	✕	◐	Later
Employee self-service + mobile	●	●	●	●	● mobile-first
Expense & travel	●	○	●	◐	Phase 2
Helpdesk / ticketing	●	◐	✕	◐	Phase 2

Module	Keka	Zoho People	greytHR	HROne	Our target (MVP)
Academic-system (SIS/timetable) integration	×	×	×	×	Phase 2

The empty column down the middle — period-level attendance, vacation-staff leave, honorarium payroll, substitution, academic integration — is the product thesis in one view.

What NONE of them do (our differentiation)

1. **Academic-calendar-driven leave** — leave cycles tied to session/term/vacation, not Jan–Dec.
2. **Vacation-staff vs non-vacation-staff logic** — teachers get vacation in lieu of EL; 12-month staff accrue EL. Drives both leave *and* pay.
3. **Period-level attendance & leave** — teacher present for 4 of 6 periods; half-day/period granularity.
4. **Substitute / relief-teacher arrangement** — auto-detect vacated periods from timetable, suggest cover, push duty, feed per-period honorarium to payroll. **No competitor has this.**
5. **Multi-scale education payroll** — parallel aided (govt/UGC pay-scale) + self-financed (management CTC) + per-lecture honorarium logic in one entity.
6. **Academic-system integration** — live SIS/timetable/exam/transport/hostel data as first-class inputs.

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> **Status: DRAFT for review.** This document lists candidate functionalities for the school HRMS, tagged MVP / Phase 2 / Later. We finalize this scope *before* writing the full technical spec. Open decisions are collected in `decisions-to-finalize.md`.

HRMS for Indian Educational Institutions — Functionality Blueprint

This blueprint defines the build scope as a set of modules. Each functionality is tagged **[MVP]** (v1), **[P2]** (Phase 2), or **[Later]**. Education-specific and India-statutory items are called out with ★ where they are the product's competitive differentiators — the things generic HRMS (Keka, Zoho People, greytHR, HROne) do NOT do.

Priority legend guiding the tagging:

- **MVP** = required to run a school's staff HR, attendance, leave and compliant payroll for at least one campus, plus the minimum education-specific hooks (academic calendar, substitution, vacation-staff leave) that make this product not-a-generic-HRMS on day one.
- **P2** = strengthens differentiation and multi-branch/group scale.
- **Later** = mature-market or nice-to-have.

PART A — CORE PLATFORM MODULES

1. Core HR / HRIS (System of Record)

Purpose: Single source of truth for all staff data, org structure, and documents that feeds attendance, leave, payroll, substitution and compliance.

- Centralized employee master (personal, job, statutory IDs — PAN/Aadhaar/UAN/ESIC/bank) **[MVP]**
- Configurable custom fields and staff-data structures **[MVP]**
- ★ **Staff-type taxonomy specific to education** (permanent PRT/TGT/PGT; contract/probationary; visiting/guest per-lecture; substitute/relief; part-time specialist — music/sports/dance; pre-primary NTT + helpers/ayahs; Class-IV support; transport; hostel; medical/counselling; coaching star-faculty; aided vs self-financed) **[MVP]**
- Org tree / reporting hierarchy and org chart **[MVP]**
- ★ **Group→Trust/Society→Campus→Department hierarchy** (multi-branch school group as first-class model, not just multi-location) **[MVP]**
- Document repository (contracts, IDs, policies) with access control **[MVP]**
- ★ **HR-controlled document governance** — HR/admin uploads employee documents captured at enrollment; **the employee cannot self-update documents.** Any change happens only through an **HR-initiated update request**, to which the employee responds by uploading; HR then **verifies/approves**. Every document has a verification status and full audit trail **[MVP]**

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- ★ **Digital service book / faculty dossier** — degrees, B.Ed, TET/CTET/NET/SET, experience certificates, police/background verification **[MVP]**
- ★ **Document-expiry alerts** — driver licences + medical fitness (transport), contract end dates (guest/contract faculty), foreign-faculty visas, background-verification validity **[MVP]**
- Confirmation / transfer / promotion / separation lifecycle tracking **[MVP]**
- Asset management and allocation (issue/return, linked to onboarding & exit) **[P2]**
- Letters & templates with mail-merge (offer, appointment, confirmation, salary revision, relieving) **[P2]**
- ★ **Qualification-to-post eligibility validation** (PRT/TGT/PGT/Principal norms; TET/NET eligibility) **[P2]**
- ★ **Pupil–teacher ratio & sanctioned-post tracking** for affiliation compliance **[P2]**
- ★ **Inspection-ready staff registers & affiliation-report exports** (CBSE bye-laws / NCTE / state board / UGC-AICTE) **[Later]**
- Announcements, directory, policy distribution & digital acknowledgement **[P2]**
- Multi-legal-entity support under one tenant **[P2]**

Who does it well: greytHR (KYE dossier, letters, 150+ registers), HROne (InboxforHR, multi-entity, digital letter ack), Keka (clean master data). *None* model the education staff taxonomy, service book, PTR or affiliation registers — that is our differentiator.

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2. Staff Attendance & Time Capture

Purpose: Tamper-proof, multi-mode attendance across scattered campuses, gates and vehicles, converted directly into payroll (LOP), leave and substitution.

- Biometric device integration (fingerprint/RFID) via push/pull API **[MVP]**
- Web/portal and mobile app punch **[MVP]**
- GPS geo-tagging + geo-fencing (enter/exit) + selfie/photo capture (anti-proxy) **[MVP]**
- Facial recognition **[P2]**
- Offline attendance capture (poor-connectivity campuses) **[P2]**
- Shift & roster management — fixed, rotating, split, flexi, weekend — for hostel/transport/security/mess staff **[MVP]**
- Auto shift allowances **[P2]**
- Attendance policies: late-coming, early-going, exceptions, weekend/holiday rules **[MVP]**
- Regularization workflow (with bulk approvals) **[MVP]**
- ★ **Half-day AND period-level attendance** (teacher present for 4 of 6 periods, not just full-day) **[MVP]**
- ★ **Timetable-linked / period-wise & subject-wise attendance** mapped to teaching workload **[P2]**
- Overtime / extra-duty auto-calculation with configurable rates **[P2]**
- Auto paid-days calculation reconciling attendance + leave + OT + holidays + weekly-offs **[MVP]**
- ★ **Multi-campus consolidated attendance** for group institutions **[MVP]**
- WhatsApp-bot / Teams punch **[Later]**

Who does it well: Keka (8 modes, 200+ devices), HROne (WhatsApp bot, real-time push API, geo-fence alerts), greytHR (Geotrack/Geomark/Geofence), Zoho (facial + IP/geo). Generic tools give full-day swipe logs only; **period-level and timetable-linked attendance is our gap-fill**.

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3. Leave Management (Academic-Calendar-Driven) ★

Purpose: Leave governed by the academic year/term/vacation/exam structure, not the Jan–Dec corporate calendar. This is one of the sharpest generic-HRMS gaps.

- Configurable leave types: EL/PL, CL, SL, Comp-off, Maternity, LOP **[MVP]**
- ★ **Education-specific types**: duty/on-duty leave (exam duty, invigilation, training, escort/competition), study leave, sabbatical, paternity/bereavement/marriage **[MVP]**
- ★ **Academic-year leave cycle** (Apr–Mar / session-based) with term & vacation calendars **[MVP]**
- ★ **Vacation-staff vs non-vacation-staff rules** — teachers get vacation in lieu of EL (EL doesn't accrue or accrues at reduced rate); non-teaching 12-month staff accrue EL **[MVP]**
- ★ **Forced vacation-leave during breaks** and vacation-period pay handling **[MVP]**
- ★ **Blackout / restricted-leave windows** around exams, admissions, result declaration **[MVP]**
- ★ **Half-day and period-level leave** application **[MVP]**
- Accrual, carry-forward caps, encashment, pro-ration rules per type **[MVP]**
- Probation/seniority-based accrual & grant rules **[P2]**
- Multi-level approval workflows with escalation **[MVP]**
- Location/campus-specific holiday calendars (National & Festival Holidays Act minimums) **[MVP]**
- ★ **Leave→substitution linkage** (approved leave triggers period-coverage need) **[MVP]**
- ★ **Leave→LOP auto-calculation** into payroll **[MVP]**
- Compensatory-off for holiday/weekend work (admissions, annual day, exams) **[MVP]**
- Leave-year-end process (close, carry-forward, new-year grant, encashment) **[P2]**
- Restricted & floating holidays, sandwich-leave policy **[P2]**
- Team/org leave calendar for managers **[P2]**
- Continuous-service tracking (gratuity 5-yr, maternity 80-day) integrated with leave/attendance ledger **[P2]**

Who does it well: greytHR & HROne (configurable engines, comp-off, sandwich/restricted-holiday, leave-year-end), Keka (highly configurable). **None model academic-year cycle, vacation-staff distinction, exam blackout, or period-level leave** — core differentiator.

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4. Payroll Engine — Multi-Scale Education Payroll ★

Purpose: Run several pay logics simultaneously (government/UGC pay-scale + management CTC + per-lecture honoraria) with full Indian statutory compliance. This and Module 5 are the product's non-negotiable strengths.

- Configurable salary structures / CTC templates (Basic, DA, HRA, conveyance, special, academic/research allowance) — formula or flat **[MVP]**
- ★ **New Labour Code compliance**: enforce excluded allowances ≤ 50% of total wages (raises PF/gratuity/bonus base) **[MVP]**
- ★ **Parallel pay groups under one entity**: aided (govt) vs self-financed (management) vs contract **[MVP]**
- ★ **7th/8th Pay Commission & UGC pay-band/grade-pay structures** with pay-matrix, auto DA, annual increment progression **[P2]** (MVP if targeting aided institutions)
- ★ **Per-lecture / per-hour / per-period honorarium engine** for visiting/guest/substitute faculty with monthly caps (e.g. UGC ₹1,500/lecture, ~₹50k cap) **[MVP]**

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- ★ **Arrears & retrospective pay-revision** processing (DA/VDA revisions, pay-commission arrears) with auto PF/ESI/TDS recompute and month-splitting **[P2]**
- Effective-dated salary revisions & increment cycles tied to academic year / DOJ / confirmation **[MVP]**
- Attendance/LOP integration & proration (mid-month join/leave, unpaid leave, part-month contract) **[MVP]**
- ★ **Overtime & extra-duty pay**: exam duty, extra classes, hostel-warden duty — configurable rates **[MVP]**
- Loans & advances with EMI schedules, interest/perquisite, auto-deduction, closure on exit **[P2]**
- Reimbursements / FBP (LTA, medical, telephone, books/academic allowance, fuel) with tax treatment **[P2]**
- Investment declaration & proof workflow (old vs new regime election) **[MVP]**
- ★ **Perquisite valuation** — rent-free/concessional accommodation for teachers, transport **[P2]**
- Off-cycle / supplementary / bonus-ex-gratia / arrears-only runs **[P2]**
- Single-click payroll run with exception/variance auto-detection before disbursement **[MVP]**
- Bank transfer / NEFT salary file generation (bank-wise/batch-wise) **[MVP]**
- Payslip generation (password-protected) + ESS download **[MVP]**
- ★ **Full & Final Settlement engine**: final salary, leave encashment (§10(10AA)), gratuity, notice pay/recovery, loan recovery, ex-gratia, TDS true-up **[P2]**
- Accounting JV/GL journal generation; Tally/SAP/Oracle export with pay-head→ledger→cost-centre mapping **[P2]**
- ★ **Multi-branch/multi-state payroll** with location-wise statutory profiles **[P2]**
- Year-end: Form 16 generation, proof reconciliation, leave carry-forward **[MVP for Form 16; rest P2]**

Who does it well: greytHR (30+ yrs, unlimited components, disbursement), Keka (compliance-first, ECR/24Q/challans/Form 16), HROne (zero-touch, both TDS regimes, 3D analytics). **None run parallel UGC-pay-scale + honorarium + management-CTC logic in one entity, nor per-lecture caps** — key differentiator.

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5. India Statutory Compliance & Filing ★

Purpose: Deep, configurable, always-current Indian statutory computation and government-portal filing. Charitable/12A status does NOT exempt these — schools must comply.

Computation (all MVP):

- **EPF** — 12% EE; ER 3.67% EPF + 8.33% EPS + ~0.50% EDLI + ~0.50% admin; wage ceiling **configurable** (₹15k → future ₹21k/25k per Jan-2026 SC direction); EPS cap ₹1,250; VPF; Code-on-Social-Security wage-base (excluded allowances ≤50%) **[MVP]**
- **ESI** — 0.75% EE + 3.25% ER on gross; ceiling ₹21k (₹25k for disability); Apr–Sep / Oct–Mar contribution periods with mid-period continuation rule **[MVP]**
- **Professional Tax** — state-configurable slabs & frequency (MH/KA monthly, TN/Kerala half-yearly, WB annual); PTEC + PTRC; not levied in Delhi/UP/Haryana/Rajasthan **[MVP]**
- **TDS on Salary (§192)** — new regime default (FY25-26 slabs, ₹75k std deduction) + old regime; perquisites; monthly challan (ITNS 281) **[MVP]**
- **Gratuity** — 15 × (Basic+DA) × years / 26; ₹20L exempt; **teachers covered** (2009 amendment, SC 2022); continuous-service tracking; provisioning **[P2]**
- **Statutory Bonus** — 8.33–20%, ₹7k calc cap, ₹21k eligibility; ★ **note educational-institution exemption (§32)** — support ex-gratia/festival bonus even where statutory bonus doesn't apply **[P2]**
- **LWF** — state- and frequency-aware (half-yearly/annual/monthly) contributions **[P2]**

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- **Minimum Wages** — validate gross/components vs current state notification + VDA by category/zone **[P2]**
- **Maternity Benefit** — 26 weeks (12 from 3rd child), 80-day eligibility, crèche obligation flag at 50+ staff **[MVP]**

Filing & Reports (statutory config keeps everything configurable):

- EPF monthly **ECR** upload + challan; UAN gen/KYC; Form 5/10 (joiners/leavers); Form 3A/6A; exit 19/10C/10D **[MVP for ECR+challan; rest P2]**
- ESI monthly challan/return; IP registration/TIC; half-yearly returns **[MVP]**
- PT state-wise challans/returns (Form IIIB etc.) **[MVP]**
- TDS quarterly **Form 24Q** (Annex I each Q; II & III in Q4) via RPU/FVU; **Form 16 Part A (TRACES) + Part B**; Form 12BA perquisites **[MVP]**
- ★ **Digitally signed Form 16 & Form 12BA [P2]**
- LWF state-wise challans/returns **[P2]**
- Gratuity computation statement + Form F (nomination) + actuarial/provision report **[P2]**
- Bonus Form C/D + ex-gratia registers **[Later]**
- Labour-Code registers: consolidated wage register, muster roll, loans/fines/deductions, leave register (Code-on-Wages/OSH formats) **[P2]**
- Reconciliation reports (PF/ESI/PT/TDS payable vs paid); statutory liability provisioning **[MVP]**
- ★ **Trust/society audit reports** — 12A/10(23C) charitable-status, Form 10B/10BB inputs, CTC-vs-actual, headcount **[P2]**
- ★ **Configurable statutory rule-set with auto-updates & alerts** (ceilings, slabs, Labour-Code effective dates by state) — architected as a versioned config layer from day one **[MVP as architecture; auto-feed P2]**

Who does it well: greytHR (Form 24Q + FVU validation, signed Form 16/12BA, multi-state PT, LWF, ECR — deepest), Keka & HROne (strong, both TDS regimes, ECR/challans). We match these AND add trust/charitable-audit + pay-commission-aware bases.

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6. ★ Substitute / Relief Teacher Management (EDUCATION-UNIQUE)

Purpose: Teacher absence is a daily safety/supervision event; generic HRMS records the leave but never arranges cover. This module is arguably the single most differentiating feature — **no competitor has it**.

- ★ **Auto-detection of vacated periods** from approved leave + live timetable **[MVP]**
- ★ **Smart substitute suggestion** ranked by subject expertise, free periods, current workload, fairness/rotation history **[MVP]**
- ★ **Daily substitution board / arrangement register** pushed to staff mobile app **[MVP]**
- ★ **Per-period / per-day substitution honorarium auto-fed to payroll [MVP]**
- ★ **Coverage-load tracking** so the same teachers aren't repeatedly burdened **[P2]**
- Manual override & swap requests **[MVP]**
- Substitution analytics (most-substituted subjects/teachers, cost) **[Later]**

Who does it well: *No competitor.* Requires timetable integration (Module 14) — the reason academic-system integration is a first-class concern.

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7. ★ Teaching Load / Workload Management (EDUCATION-UNIQUE)

Purpose: A teacher's presence and pay tie to periods taught and prescribed teaching-hour norms (e.g. CBSE ~1200 hrs/yr) — a generic time-clock cannot capture this.

- ★ **Periods-per-week / per-day load per teacher** with min–max target ranges **[P2]**
- ★ **Subject–class–section allocation** and workload balancing across faculty **[P2]**
- ★ **Free-period & non-teaching-duty visibility** for fair distribution **[P2]**
- ★ **Overload / extra-period tracking → extra-remuneration in payroll [P2]**
- ★ **Compliance reporting vs mandated teaching-hour norms [Later]**

Who does it well: *No competitor.* Consumes timetable/SIS data.

---

8. Onboarding & Offboarding (Seasonal/Academic)

Purpose: Joiner–mover–leaver lifecycle tuned to seasonal, admissions-driven, academic-year-contract hiring.

- Digital pre-boarding with secure document-upload links **[MVP]**
- Onboarding workflows, task checklists, induction **[MVP]**
- ★ **Bulk/seasonal onboarding** with fixed academic-year contracts + auto-expiry **[MVP]**
- ★ **Admissions-season temp staff** (counsellors, tele-callers, exam invigilators) **[P2]**
- ★ **Mass renewal / non-renewal workflow at session end [MVP]**
- Confirmation & probation tracking **[MVP]**
- Buddy assignment **[Later]**
- Inter-branch transfer with service continuity (see Module 12) **[P2]**
- Exit/offboarding: clearance checklist, class handover, F&F linkage, access revocation, exit interview **[P2]**
- E-signature on offer/appointment/relieving letters **[P2]**

Who does it well: HROne & greytHR (checklists, F&F linkage, pre-boarding links), Keka (ATS→onboarding). Seasonal bulk/academic-contract auto-expiry is our add.

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9. Employee Self-Service (ESS) & Staff Mobile App

Purpose: Phone-first for a classroom-bound, campus-scattered workforce that is rarely at a desk. Mobile is not optional — it's primary.

- Mobile leave apply/approve + balance view **[MVP]**
- Attendance mark-in (geo/selfie) + regularization on mobile **[MVP]**
- Payslip, Form 16, tax-slip self-download **[MVP]**
- Investment declaration + proof upload **[MVP]**
- ★ **Daily substitution/arrangement duty notifications [MVP]**
- ★ **Timetable view on the same app used for academics [P2]**
- Profile/document self-update **[MVP]**
- Circular/notice + document-request workflows **[P2]**
- Manager self-service (approvals, team views) **[MVP]**
- Native iOS/Android apps **[MVP]**
- HR chatbot (Bella/Zia/One-AI style) **[Later]**

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- WhatsApp/Teams bot actions **[Later]**

Who does it well: HROne (110+ tasks, award-winning app), greytHR (mobile-first ESS, Bella), Keka & Zoho (solid apps). Substitution-duty push notification is education-specific.

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10. Faculty Performance Appraisal & Professional Development ★

Purpose: Education appraisal uses classroom observation, student/parent feedback, result linkage and UGC API scoring — not corporate KRA/OKR.

- Configurable review cycles (term/annual) **[P2]**
- ★ **Classroom-observation & peer-review workflows [P2]**
- ★ **Student result / outcome linkage** into appraisal **[P2]**
- ★ **Student/parent 360-degree feedback [P2]**
- ★ **UGC API / academic self-appraisal scoring** for college faculty **[Later]**
- ★ **CPD / in-service training-hours tracking & certification [P2]**
- Confirmation/promotion/pay-scale-progression recommendations **[P2]**
- Standard goal-setting, self/manager review, 360 (corporate baseline) **[P2]**

Who does it well: Keka (mature OKR/360/calibration), HROne (9-box/succession), Zoho & greytHR (appraisal cycles). All corporate-shaped; **classroom observation, result-linkage, parent 360, UGC API are our differentiators.**

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11. Recruitment / ATS ✅ NOW MVP (scope change 2026-07-15)

Purpose: Hiring teaching and admin staff, with seasonal surges and mandated-qualification screening. **Promoted into Phase 1** — the hire-to-onboard pipeline feeds Core HR directly, so building it in MVP avoids double data entry and gives a complete joiner journey from day one.

- Requisition with budget-based approval routing **[MVP]**
- Career portal + job posting/syndication **[MVP]**
- Resume capture & candidate pipeline (stages, kanban) **[MVP]**
- ★ **Qualification/eligibility screening** (TET/CTET/NET/B.Ed) at shortlist **[MVP]**
- Interview scheduling + structured feedback/scorecards **[MVP]**
- Offer letter with CTC calculator + digital acceptance **[MVP]**
- ★ **Candidate→Employee auto-conversion** (carries captured documents into the staff record) **[MVP]**
- Background/police verification tracking (child-safety mandate) **[MVP]**
- Resume parsing (AI) **[P2]**
- Recruitment analytics (time-to-hire, source, funnel) **[P2]**
- ★ **Coaching-centre commission/incentive-linked faculty & counsellor pay** hooks **[Later]**

Who does it well: HROne (BGV vendors, requisition budgets), greytHR Recruit (AI parsing, Lisa bot), Keka Hire. All are add-ons; we bundle qualification-screening and make candidate→employee conversion carry documents into the new document-governance model (Module 1).

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12. ★ Multi-Branch / Trust Group Consolidation (EDUCATION-UNIQUE FRAMING)

Purpose: School groups/societies/trusts run many campuses under one entity needing centralized policy with campus-level control — a single-company HRMS can't model this.

- ★ **Group→campus hierarchy** with centralized policy + localized execution **[MVP]**
- ★ **Inter-branch employee transfer** with service continuity & pay carry-over **[P2]**
- ★ **Consolidated cross-campus payroll, PF/ESI & compliance reporting [P2]**
- ★ **Trust/society-aligned (12A/80G) fund & academic-year accounting integration [P2]**
- ★ **Campus-wise + group-wide leadership dashboards [P2]**
- Aided vs self-financed dual-regime reporting within one campus **[P2]**

Who does it well: Keka/HROne (multi-legal-entity/multi-location) partially. The **trust-group hierarchy + service-continuity transfers + 12A/80G accounting** framing is ours.

---

13. Expense, Reimbursement & Travel

Purpose: Policy-driven claims linked to payroll.

- Mobile expense/claim submission with receipt upload **[P2]**
- Configurable expense policies by category/role **[P2]**
- Multi-level approval workflows **[P2]**
- Reimbursement payout via payroll **[P2]**
- Travel request → expense auto-pull **[Later]**
- Mileage/distance reimbursement for transport/field staff **[Later]**

Who does it well: Keka (OCR, travel desk), HROne, greytHR (payroll-integrated claims), Zoho (via Zoho Expense).

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14. ★ Academic-System Integration Layer (STRATEGIC)

Purpose: The HRMS must not live in a silo — substitution, workload, exam duty and attendance depend on live academic data. This is what turns four generic-HRMS gaps into features.

- ★ **School ERP / SIS connector** (Fedena, Teachmint, myClassCampus, Entab, MasterSoft, edumerge) — staff master, class-section-subject structure, academic calendar **[P2, but design API from MVP]**
- ★ **Timetable/scheduling engine integration** — derive teaching load, free periods, exact periods needing substitution **[P2]** (MVP-critical if substitution ships in v1 — minimum: import timetable)
- ★ **Examination/result system** — exam-duty (invigilation/evaluation) allocation, duty-leave, honorarium, result-linked appraisal **[P2]**
- ★ **Transport/fleet** — driver-conductor rostering, route duty, licence/medical-fitness expiry **[P2]**
- ★ **Hostel/boarding** — warden/matron/mess shift rosters **[P2]**
- ★ **Admissions/CRM** — trigger seasonal hiring & counsellor incentive pay **[Later]**
- LMS/digital classroom — correlate online-class hours with workload/appraisal **[Later]**
- Government/regulatory portals — CBSE/ICSE/state affiliation, UGC/AICTE, aided-school treasury/DDO **[Later]**

Who does it well: *No generic HRMS integrates academic systems* — this is the entire product thesis.

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15. Supporting Modules (Analytics, Helpdesk, Engagement, Workflows, Integrations, Documents)

HR Analytics & Reporting

- Standard payroll/attendance/leave/compliance reports (audit-ready) **[MVP]**
- Campus-wise & group dashboards (headcount, attrition, payroll cost, statutory liability) **[P2]**
- Custom report/query builder **[P2]**
- ★ **PTR, teaching-hour-norm, sanctioned-post compliance reports** [P2]

HR Helpdesk / Ticketing

- Ticket categories, auto-assignment, SLA/TAT tracking, grievance handling **[P2]**

Engagement & Surveys

- Pulse/eNPS surveys, recognition/badges, bulletin board, birthday/anniversary greetings **[Later]**

Workflows & Automation

- Configurable multi-level approval chains across leave/attendance/payroll/expense/regularization **[MVP]**
- Rule-based routing, notifications, escalations **[MVP]**
- Payroll scheduler, auto-regularization, leave-accrual automation **[P2]**

Integrations & API

- Biometric device families (ZKTeco/eSSL/Matrix/Realtime/Spectra) **[MVP]**
- Tally / accounting export **[P2]**
- Open REST API + webhooks **[P2]**
- SMS/WhatsApp/email communication layer (leave approvals, substitution alerts, payslips, circulars) **[MVP for SMS/email; WhatsApp P2]**
- SSO/SAML, Aadhaar AGBAS **[Later]**

LMS / Staff Training — staff CPD/in-service training management (not student LMS) **[Later]**

Who does it well: greytHR (150+ reports, Tally/Zoho Books), HROne (Tally/SAP/Oracle, 200+ devices, REST), Keka (marketplace), Zoho (suite integration).

PART B — CROSS-CUTTING CONCERNS

1. Roles/Permissions & Multi-Branch Hierarchy

- RBAC with role- and department-based data-visibility rules **[MVP]**
- ★ **Scoped access by group/trust/campus level** — group admin, campus principal, department head, HR/payroll operator, accountant, employee, manager **[MVP]**
- Field-level PII visibility rules (bank/PAN/Aadhaar/salary) **[MVP]**
- Aided-vs-self-financed data-segment permissions **[P2]**
- Maker-checker on payroll & statutory runs **[P2]**

2. Employee Lifecycle States

- Pre-boarding → Onboarding → Probation → Confirmed → Active → On-leave/Sabbatical → Transferred → Notice → Exited/Relieved → F&F-closed **[MVP]**
- ★ **Contract/visiting states with auto-expiry** and mass renewal/non-renewal at session end **[MVP]**

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- ★ **Vacation-staff vs non-vacation-staff attribute** driving leave/pay behavior **[MVP]**
- Service-continuity flag across inter-branch transfers **[P2]**

3. Mobile / ESS — phone-first is primary UI (see Module 9); offline tolerance for weak-connectivity campuses **[MVP]**

4. Notifications — multi-channel (in-app, push, SMS, email; WhatsApp later): approvals, substitution duty, document-expiry alerts, statutory due-dates, payslip release, contract expiry **[MVP for core channels]**

5. Audit & Traceability — immutable audit log on payroll runs, statutory filings, master-data changes, approvals; versioned statutory-config history; who-changed-what-when **[MVP for payroll/statutory; full P2]**

6. Data Security & PII — encryption at rest/in transit; password-protected payslips/Form 16; PII field-masking; consent & retention aligned to DPDP Act; ★ **child-safety context** (background-verification records, restricted access to staff-of-minors data); role-scoped exports **[MVP]**

7. Integrations — architected around a stable internal staff/attendance/timetable API so SIS, biometric, timetable, payroll-accounting and communication connectors plug in without core changes **[MVP as architecture]**

PART C — RECOMMENDED MVP SCOPE SUMMARY

Thesis for v1: Ship a *compliant Indian staff-payroll HRMS that is unmistakably built for schools* — i.e. match greytHR/Keka on statutory payroll for one-to-few campuses, and lead with the four education gaps competitors cannot close: academic-calendar leave, vacation-staff logic, period-level attendance/leave, and substitute-teacher arrangement.

MVP MUST include:

1. **Core HR** — staff master with **education staff-type taxonomy**, group→trust→campus hierarchy, service book, **document-expiry alerts**, and **HR-controlled document governance** (HR uploads at enrollment; employee updates only via an HR-initiated request → HR verification; no self-update).
2. **Attendance** — biometric + geo/selfie mobile, shifts/rosters, **half-day & period-level**, regularization, auto paid-days, multi-campus consolidation.
3. **Leave** — **academic-year cycle, vacation vs non-vacation staff, exam/admission blackouts, period-level leave, leave→substitution & leave→LOP linkage**, education leave types, holiday calendars, approvals.
4. **Payroll** — configurable structures with **50%-allowance Labour-Code rule, parallel pay groups (aided/self-financed/contract), per-lecture honorarium engine with caps**, LOP proration, extra-duty pay, investment declaration, single-click run with variance detection, bank file, payslips, Form 16.
5. **Statutory — FULL greytHR parity (locked D3):** PF (ECR + challan, configurable ceiling), ESI, PT (state-configurable), TDS §192 (both regimes) + **Form 24Q eTDS returns + Form 16 (Part A & B) + Form 12BA, digitally signed**, **gratuity (with continuous-service tracking), statutory bonus, LWF, minimum-wage validation**, Maternity; reconciliation & liability-provisioning reports; **versioned statutory-config layer**.
6. **Full & Final Settlement engine** — MVP-coupled to full statutory parity: final salary, **leave encashment (§10(10AA)), gratuity payout**, notice pay/recovery, loan recovery, TDS true-up.
7. **Onboarding** — with **seasonal bulk + academic-year contract auto-expiry + session-end renewal**.
8. **Recruitment / ATS** *(promoted to MVP, 2026-07-15)* — requisition + approval, job posting/career page, candidate pipeline, **qualification/eligibility screening**, interview scheduling + scorecards, offer + digital

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acceptance, **candidate→employee conversion carrying documents into HR-governed storage**, BGV/police-verification tracking.

9. **ESS & mobile app** — leave, attendance, payslip/Form 16, investment declaration, manager approvals, **and responding to HR document-update requests** (no unilateral self-update). (Substitution-duty notifications move to Phase 2.)
10. **Cross-cutting** — RBAC with campus-scoped access, lifecycle states (incl. contract auto-expiry & vacation-staff attribute), multi-channel notifications, payroll/statutory audit log, PII security, biometric + SMS/email integration APIs.

> **Locked-scope deltas:** *(2026-07-14)* Multi-branch **group→trust→campus hierarchy is IN MVP** (D1: multi-location K-12). **Substitute-teacher management is DEFERRED to Phase 2** (D2) — and with it **period-wise attendance/leave, timetable import, and teaching-load/workload**. Half-day granularity stays MVP. Full statutory parity + F&F are pulled INTO MVP (D3). *(2026-07-15)* **Recruitment/ATS is promoted INTO MVP**; **HR-controlled document governance** (HR-upload + request-gated employee updates) is added to Core HR/ESS.

Explicitly DEFERRED from MVP (Phase 2): ★ substitute-teacher management, period-level attendance/leave, timetable import, teaching-load balancing (D2); full pay-commission/UGC pay-matrix engine; arrears/retro pay-revision engine; faculty appraisal (classroom-observation/result-linkage/parent-360); AI resume parsing & recruitment analytics *(core ATS is now MVP)*; expense/travel; helpdesk; inter-branch transfers with service continuity; deep SIS/exam/transport/hostel integrations; trust 12A/80G accounting; custom report builder.

Later: UGC API scoring, affiliation-inspection exports, engagement/surveys, staff LMS/CPD, WhatsApp bot, admissions/CRM & govt-portal integrations, AI assistants.

Guiding build principles:

- Keep **all statutory bases, ceilings, slabs and effective dates configurable/versioned** from day one — the Code on Social Security (in force 21-Nov-2025), the pending PF-ceiling revision, and state-wise PT/LWF/min-wage variance make hard-coding fatal.
- Design the **timetable/SIS/attendance internal API before** substitution and workload — those education features are worthless without live academic data.
- Treat **mobile as the primary client**, not a companion — the teaching workforce is classroom-bound and desk-less.
- The **vacation-staff vs non-vacation-staff attribute and the academic-year calendar are foundational data model decisions** that ripple through leave, attendance and payroll — get them into the schema at MVP, not bolted on later.

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Development Specification — School HRMS

Audience & purpose. This is the authoritative build document for engineering the MVP. It is self-contained: architecture, the full data model, per-module functional + technical specifications, workflows, the API surface, security, non-functional requirements, testing, and a sprint-by-sprint build plan. A business reader can follow Parts I and III's "Purpose" and workflow narratives; engineers should read everything.

This document supersedes the earlier `foundation-and-workflows.md` (its content is folded in and extended here). Scope is governed by `functionality-blueprint.md` (MVP / Phase 2 / Later) and the locked decisions in `decisions-to-finalize.md`.

Recent scope changes (2026-07-15): (1) **Recruitment/ATS is in the MVP.** (2) **HR-controlled document governance** — HR uploads staff documents at enrollment; employees cannot self-update; changes flow only through an HR-initiated request → employee upload → HR verification.

PART I — OVERVIEW & ARCHITECTURE

1. Product scope (MVP)

A compliance-grade Indian staff-payroll HRMS **built for schools**, targeting **private, multi-branch K-12 groups** first, delivered as **multi-tenant SaaS** with a **single-tenant on-premise** option for enterprise.

MVP modules (full detail in Part III):

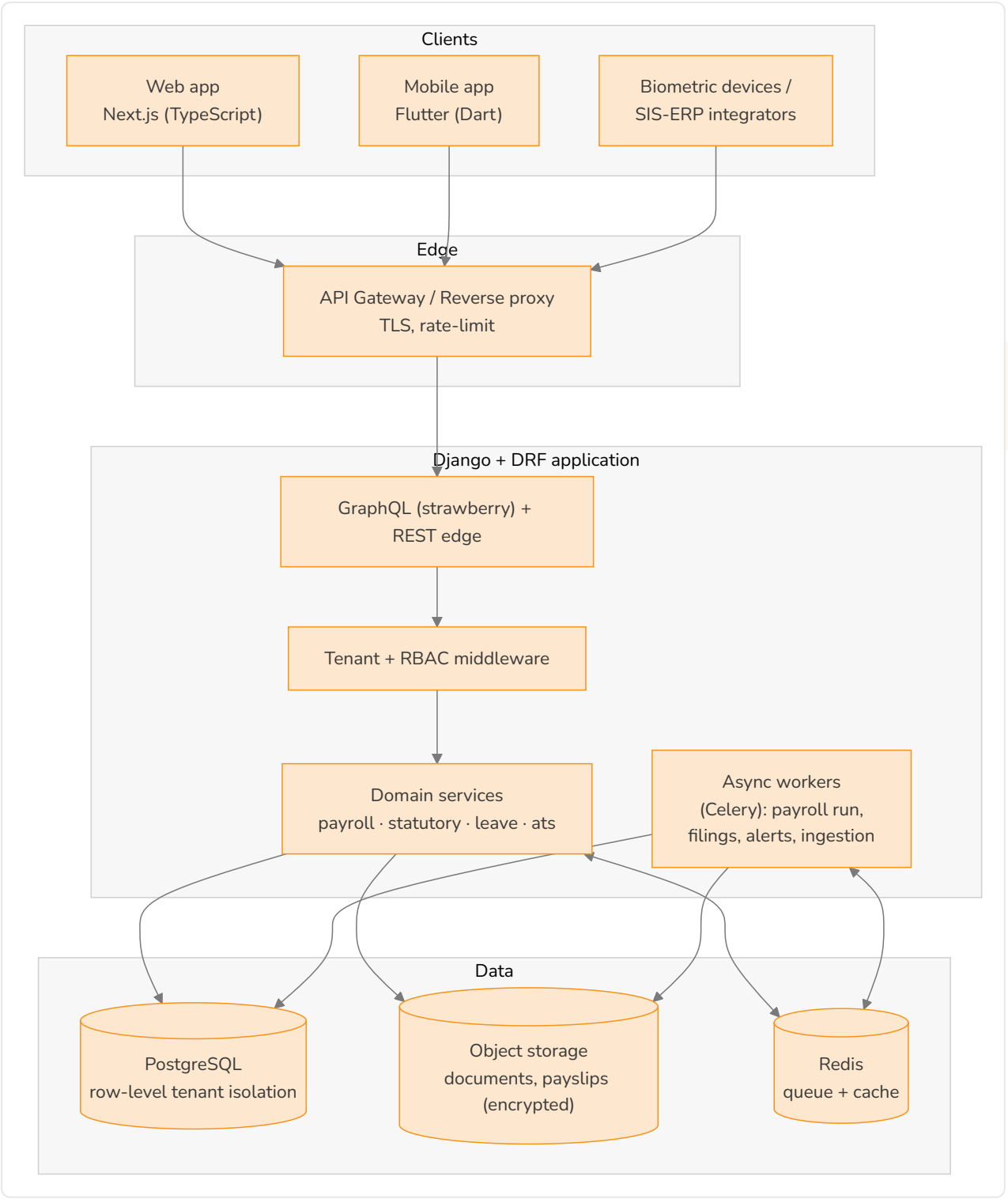
#	Module	One-line
M1	Organization & Tenancy	Tenant → group → trust → campus → department; the isolation boundary
M2	Identity, Roles & Access	Login, RBAC = permission × scope (group/campus/department)
M3	Core HR — People	Staff master, education staff taxonomy, contracts, vacation-staff flag, service book
M4	Document Management & Verification	HR-uploaded, request-gated employee updates, verification, expiry alerts
M5	Recruitment / ATS	Requisition → posting → pipeline → interview → offer → convert to employee
M6	Onboarding	Seasonal/academic-year contracts, checklists, candidate carry-over
M7	Academic Calendar	Academic year, terms, vacation periods, holiday calendars
M8	Leave	Academic-calendar cycle, vacation-staff logic, approvals, LOP linkage

#	Module	One-line
M9	Attendance	Biometric + geo/selfie mobile, shifts, half-day, paid-days reconciliation
M10	Payroll	Multi-scale (CTC + honorarium), LOP proration, variance review, bank file
M11	Statutory Compliance	PF/ESI/PT/TDS/gratuity/bonus/LWF + filings (ECR, 24Q, Form 16)
M12	Full & Final Settlement	Leave encashment, gratuity, recoveries, TDS true-up
M13	Employee Self-Service & Mobile	Phone-first: leave, attendance, payslip, doc-request responses
M14	Notifications & Alerts	Multi-channel: approvals, expiries, statutory due dates
M15	Reporting & Platform	Standard reports, audit log, dashboards
M16	Integrations	Biometric devices, SMS/email, accounting export, webhooks

Explicitly Phase 2+ (schema-ready, not built): substitute-teacher arrangement, period-level attendance/leave, timetable import, teaching-load, faculty appraisal, pay-commission/UGC engine, expense/travel, helpdesk, deep SIS integration. See the roadmap in `functionality-blueprint.md`.

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2. System architecture



Stack (locked):

Concern	Choice	Rationale
Backend	**Django + Django REST Framework**	Batteries-included (admin, auth, ORM, migrations); ideal for a small team
Database	**PostgreSQL**	Row-level security for tenant isolation; JSONB for versioned statutory config; strong integrity

Concern	Choice	Rationale
Async	**Celery + Redis**	Payroll runs, statutory filing generation, device ingestion, scheduled alerts
Object storage	S3-compatible (SaaS) / local encrypted volume (on-prem)	Documents, payslips, Form 16 — encrypted at rest
Web	**Next.js (React/TypeScript)**	SSR-capable admin/HR console
Mobile	**Flutter (Dart)**	One codebase, phone-first ESS
API layer	**strawberry-django GraphQL (primary) + DRF REST edge**	GraphQL SDL + OpenAPI both → generated TypeScript + Dart clients
Auth	JWT (access + refresh)	Stateless; works across web + mobile

3. Multi-tenancy & deployment

- **Tenant = one customer account (SaaS) or one whole install (on-prem).** The tenant boundary sits **above** the `group → trust → campus → department` hierarchy.
- **Isolation:** every table (except `Tenant`) carries `tenant_id`. Enforcement is layered:
  1. A Django base model + manager that auto-filters by the request's tenant.
  2. Request middleware resolves the tenant (from JWT claim / subdomain) and sets a Postgres session var.
  3. **PostgreSQL Row-Level Security** policies keyed on that session var as a defence-in-depth backstop — even a query that forgets the filter cannot cross tenants.
- **On-prem = the identical image** deployed with a single tenant (`deployment_mode='on_prem'`); no code fork.
- **Scaling** is per-tenant/per-campus; the largest single group sets the ceiling, not the SaaS population.

4. Cross-cutting foundations

These are built **first** and every module depends on them.

4.1 RBAC — permission × scope

Authorization is two-dimensional: **what** (a permission like `leave.approve`) × **where** (a scope: group / campus / department + id). A `RoleAssignment` binds a user to a role at a scope. Every API request is filtered to the union of the caller's scoped assignments. **Field-level PII permissions** (bank, PAN, Aadhaar, salary) are separate from record access.

4.2 Audit — immutable trail

Every mutating action writes an `AuditLog` (actor, action, entity, before/after, timestamp). Mandatory and non-bypassable for payroll runs, statutory filings, approvals, document changes, and master-data edits.

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4.3 Effective-dating

Records that change over time and must retain history (salary, statutory config, contracts) are **never edited in place** — a new dated row supersedes the old. One consistent pattern, reused everywhere.

4.4 Versioned statutory config

`StatutoryConfig` is an immutable, versioned snapshot of all rates/ceilings/slabs. Each `PayrollRun` pins the version it used, so a mid-year rule change never corrupts past payrolls. Statutory numbers live in data, never in code.

4.5 Document governance principle

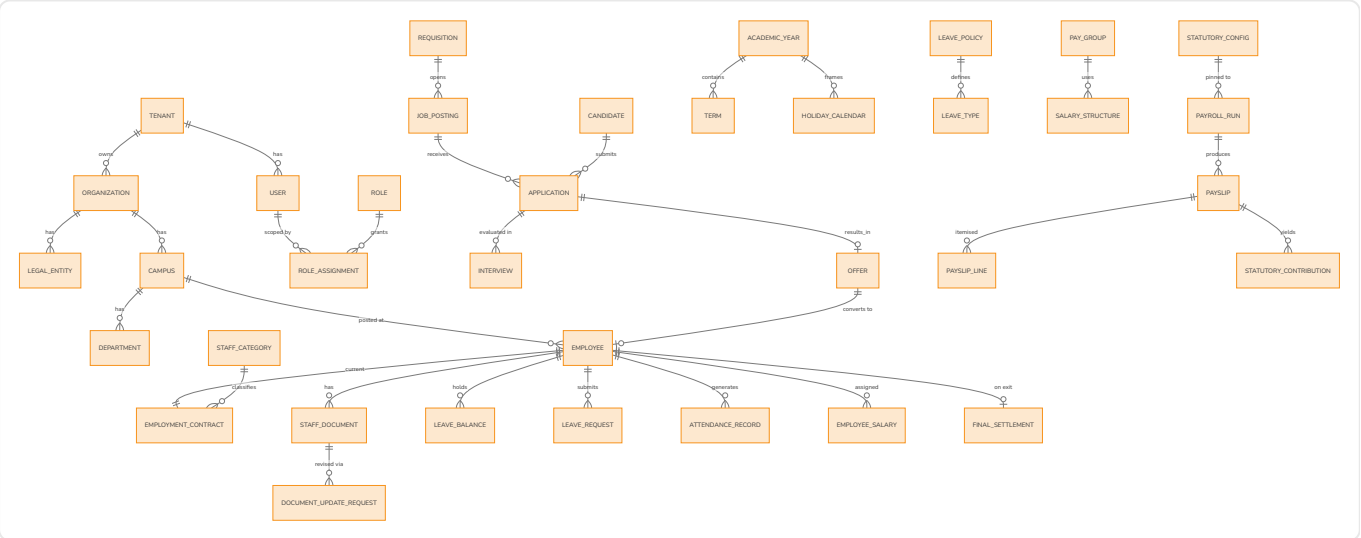
Employee documents are **HR-owned**. HR uploads at enrollment (or they arrive via ATS conversion). **Employees have no direct write path** to their documents; the only way an employee supplies/updates a document is by responding to an **HR-initiated** `DocumentUpdateRequest`, after which HR **verifies**. This is enforced at the API/permission layer, not just the UI. Full detail in Module M4.

4.6 Notifications

A channel-agnostic `Notification` service (in-app, push, SMS, email; WhatsApp later) used by approvals, document requests, expiry alerts, statutory due dates, and payslip release.

PART II — DATA MODEL

5. Entity-relationship overview



6. Entity catalog

Every entity carries `id`, `tenant_id`, `created_at`, `updated_at` (omitted below for brevity). Enumerated values shown in `code(...)`.

6.1 Tenancy & organization

Entity	Key fields	Notes
Tenant	name, deployment_mode `(saas/on_prem)`, status, data_region	Isolation boundary; no `tenant_id` on itself
Organization	org_id, name	The group/society
LegalEntity	org_id, name, registration_type `(trust/society/sec8)`, pan, tan, pf_estab_id, esi_code, reg_12a	Employer of record; statutory returns file here
Campus	org_id, legal_entity_id, name, address, **state**, timezone	State drives PT/LWF/min-wage
Department	campus_id, name, type `(academic/admin/transport/hostel/…)`	

6.2 Identity & access

Entity	Key fields	Notes
User	email/phone, password_hash, employee_id (nullable), is_active, mfa	Login identity
Role	name, permission_set (JSON), is_system	e.g. GroupAdmin, Principal, HROperator, Employee
RoleAssignment	user_id, role_id, scope_level `(group/campus/department)`, scope_id	Two-dimensional RBAC binding

6.3 People

Entity	Key fields	Notes
Employee	campus_id, department_id, emp_code, name, dob, gender, contact, **pan, aadhaar(masked), uan, esic_no, bank(enc)**, photo, status	PII encrypted + field-permissioned
StaffCategory	name, kind `(teaching/non_teaching/support/guest)`, min_qualification	Drives eligibility
EmploymentContract	employee_id, staff_category_id, employment_type `(permanent/probation/contract/visiting)`, **is_vacation_staff**, doj, confirmation_date, contract_end_date, pay_group_id, notice_period, status	Center of gravity — vacation flag + auto-expiry
ServiceRecord	employee_id, event_type `(joined/confirmed/promoted/transferred/revised/exited)`, effective_date, details	Immutable career timeline; feeds gratuity

6.4 Documents & verification (M4)

Entity	Key fields	Notes
StaffDocument	employee_id, doc_type `(degree/bed/tet/police_verif/license/aadhaar/pan/…)`, file_ref(enc), issued_on, **expires_on**, **verification_status** `(pending/verified/rejected)`, uploaded_by_user_id, source `(hr_upload/ats_import/request_response)`	HR-owned; employees never write directly
DocumentUpdateRequest	employee_id, doc_type, reason, requested_by_user_id (HR/admin), status `(open/submitted/verified/rejected/cancelled)`, due_date, response_file_ref, verified_by_user_id, notes	**The only channel** by which an employee supplies/updates a document

6.5 Recruitment / ATS (M5)

Entity	Key fields	Notes
Requisition	campus_id, department_id, staff_category_id, title, headcount, budget_ctc, justification, status `(draft/pending_approval/approved/rejected/closed)`, approval_chain (JSON)	Budget-based approval routing
JobPosting	requisition_id, channel `(career_site/board/referral)`, published_at, closes_at	
Candidate	name, email, phone, resume_ref, **qualifications (JSON)**, source	De-duplicated talent pool
Application	candidate_id, job_posting_id, stage `(applied/screening/shortlisted/interview/offer/hired/rejected)`, **eligibility_flags (JSON)**, notes	Pipeline; eligibility = TET/CTET/NET/B.Ed screening
Interview	application_id, round, scheduled_at, panel (JSON), **scorecard (JSON)**, outcome	
Offer	application_id, ctc, pay_group_id, join_date, status `(draft/sent/accepted/declined/expired)`, letter_ref, accepted_at	Digital acceptance
(conversion)	Offer `accepted` → creates **Employee** + **EmploymentContract**, and **carries the candidate's documents into StaffDocument** (source= `ats_import`, verification_status= `pending`)	Feeds M4 governance

6.6 Academic foundation (M7)

Entity	Key fields	Notes
AcademicYear	scope (campus/org), label, start_date, end_date, is_current	Leave reference frame
Term	academic_year_id, name, start_date, end_date, type `(teaching/exam/vacation)`	Vacation terms drive teacher leave
HolidayCalendar	campus_id, academic_year_id, name	Per-campus

Entity	Key fields	Notes
Holiday	calendar_id, date, name, type `(national/festival/restricted/floating)`	

6.7 Leave (M8)

Entity	Key fields	Notes
LeavePolicy	scope, academic_year_id, name	Tied to an academic year
LeaveType	policy_id, code `(EL/CL/SL/COMP/DUTY/MAT/LOP)`, accrual_rule (JSON), carry_forward_cap, encashable, **applies_to_vacation_staff**, allows_half_day	Vacation-staff-aware
LeaveBalance	employee_id, leave_type_id, academic_year_id, opening, accrued, used, balance	Per-AY ledger
LeaveRequest	employee_id, leave_type_id, from_date, to_date, is_half_day, day_count, reason, status, approval_chain (JSON)	Updates balance + LOP

6.8 Attendance (M9)

Entity	Key fields	Notes
Shift	campus_id, name, start/end, type `(fixed/flexi/rotating)`, grace_minutes	Rosters
AttendanceRecord	employee_id, date, shift_id, first_in, last_out, source `(biometric/mobile/web)`, geo, selfie_ref, status `(present/absent/half_day/leave/holiday/weekly_off)`, worked_hours	One/day; half-day in MVP
Regularization	attendance_id, requested_status, reason, approval_chain, status	Corrections

6.9 Payroll (M10)

Entity	Key fields	Notes
PayGroup	legal_entity_id, name, kind `(ctc_monthly/contract/honorarium)`	Parallel pay logics
SalaryStructure	pay_group_id, name, effective_from	Component template
PayComponent	structure_id, code `(BASIC/DA/HRA/SPL/…)`, calc_type `(flat/percent/formula)`, value, taxable, pf_applicable, esi_applicable	Enforces 50%-allowance rule
EmployeeSalary	employee_id, structure_id, ctc, effective_from	Effective-dated; revisions add rows
HonorariumRule	pay_group_id, unit `(lecture/period/day)`, rate, monthly_cap	Guest/substitute pay
PayrollRun	legal_entity_id, period, status `(draft/variance_review/approved/disbursed)`, statutory_config_id	Pins config version
Payslip	payroll_run_id, employee_id, paid_days, gross, deductions, net, is_locked	Password-protected PDF

Entity	Key fields	Notes
PayslipLine	payslip_id, component_code, amount, type `(earning/deduction/employer_contrib)`	Itemised

6.10 Statutory (M11)

Entity	Key fields	Notes
StatutoryConfig	version, effective_from, rules (JSON: PF rate+ceiling, EPS cap, ESI rate+ceiling+periods, PT slabs per state, TDS slabs old/new, gratuity formula, bonus %/caps, LWF per state, min-wage)	Immutable, versioned
StatutoryContribution	payslip_id, kind `(pf/esi/pt/tds/lwf)`, employee_amt, employer_amt, base_amount	Rolls up into filings
StatutoryFiling	legal_entity_id, kind `(ecr/esi_return/pt_challan/form_24q/form_16)`, period, artifact_ref, status	Government artefacts
InvestmentDeclaration	employee_id, financial_year, regime `(old/new)`, declarations (JSON), proofs	Drives TDS §192

6.11 Exit & platform (M12, M14, M15)

Entity	Key fields	Notes
FinalSettlement	employee_id, last_working_day, leave_encashment, gratuity, notice_recovery, loan_recovery, tds_trueup, net_payable, status	F&F engine
Notification	recipient, channel `(in_app/push/sms/email)`, template, payload, status	Multi-channel
AuditLog	actor_user_id, action, entity, entity_id, before/after (JSON), timestamp	Immutable, read-only

PART III — MODULE SPECIFICATIONS

Each module follows the same template: **Purpose · Actors · Entities · Business rules · Workflow(s) · API · Permissions · Acceptance criteria.**

M1 — Organization & Tenancy

Purpose: Model a customer's structure and enforce isolation. **Actors:** Group Admin (setup), platform (isolation). **Entities:** Tenant, Organization, LegalEntity, Campus, Department. **Business rules:** every record is tenant-scoped; campus carries `state` (statutory driver); a legal entity groups campuses for payroll/filing; on-prem = single tenant. **API:** `GET/POST/PATCH …/organizations|legal-entities|campuses|departments`. **Permissions:** `org.manage` at group scope (Group Admin only). **Acceptance:** a Group Admin can create a group with ≥2 campuses in different states; no query returns another tenant's rows (verified by an RLS cross-tenant test).

M2 — Identity, Roles & Access

Purpose: Authentication + two-dimensional authorization. **Actors:** all users; Group/Campus admins manage roles. **Entities:** User, Role, RoleAssignment. **Business rules:** permission × scope on every request; field-level PII permissions distinct from record access; JWT access+refresh; optional MFA for admin/payroll roles; maker-checker capability flag on roles. **Seed roles:** Group Admin, Campus Principal, HR Operator, Payroll Operator, Accountant, Department Head, Employee. **API:** `POST /auth/login|refresh`, `GET /me`, `…/roles`, `…/role-assignments`. **Permissions:** `iam.manage` (group/campus scope). **Acceptance:** a Principal scoped to Campus A cannot read Campus B data; a Department Head can approve leave but cannot see bank details unless granted the PII permission.

M3 — Core HR (People)

Purpose: The staff system of record. **Actors:** HR Operator, Principal, Employee (read-only self). **Entities:** Employee, StaffCategory, EmploymentContract, ServiceRecord. **Business rules:** `is_vacation_staff` and `contract_end_date` live on the contract; salary/leave/attendance all derive behaviour from the contract; PII encrypted; every lifecycle change writes a ServiceRecord. **Workflow — teacher onboarding (also see M6):** create Employee → contract (category, vacation flag, end-date) → salary assignment → leave-balance init → user invite → ServiceRecord `joined`. **API:** `…/employees`, `…/employees/{id}/contracts`, `…/staff-categories`. **Permissions:** `people.manage` (campus scope); `people.read` for managers; PII gated by `people.pii.read`. **Acceptance:** onboarding a PGT with `is_vacation_staff=true` initialises leave balances that exclude/reduce EL.

M4 — Document Management & Verification ★ (governance model)

Purpose: Keep staff documents **HR-owned and verified**, with a controlled channel for employee updates. **Actors:** HR Operator/Admin (upload, request, verify), Employee (respond only).

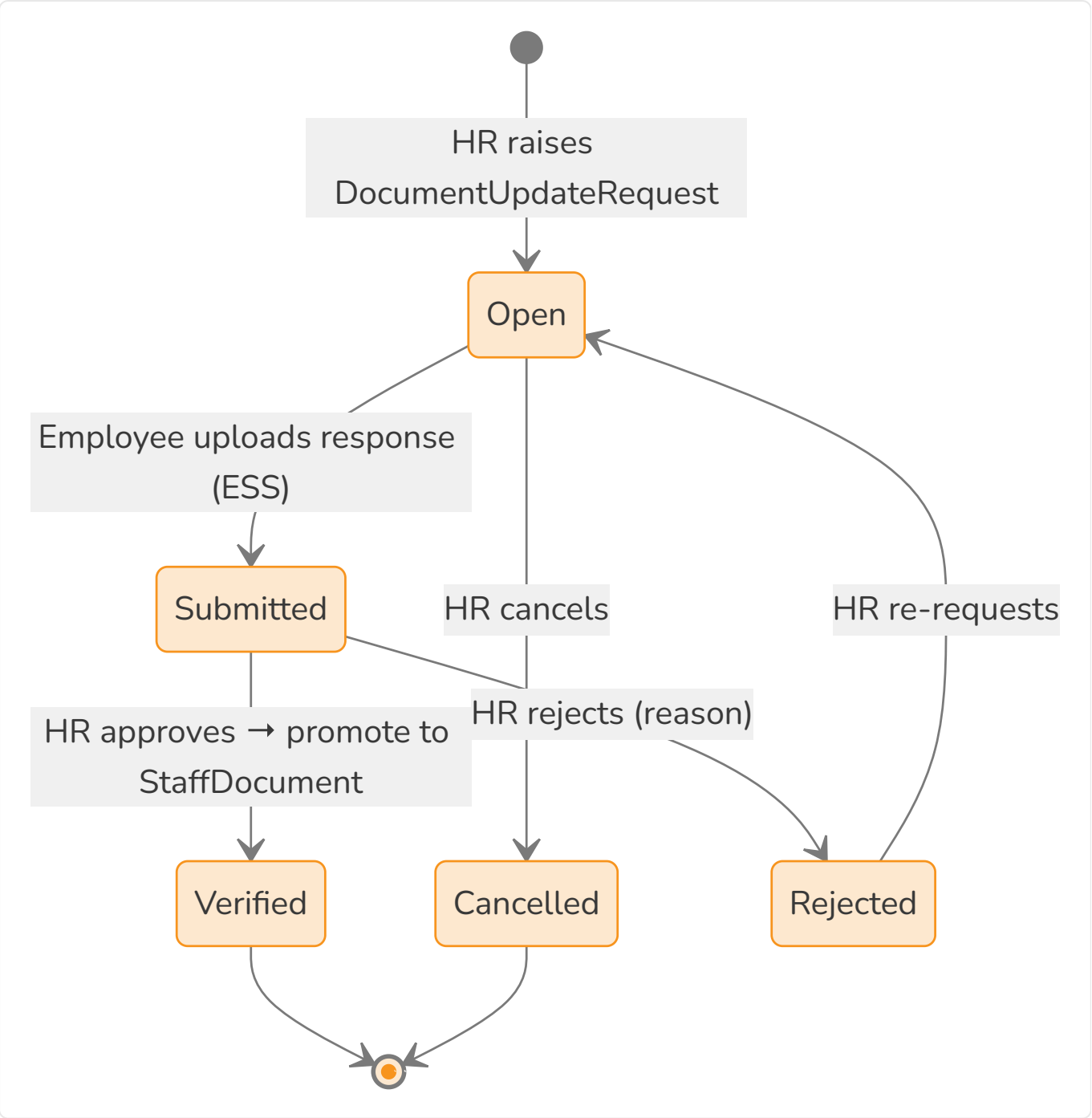
Entities: StaffDocument, DocumentUpdateRequest.

Business rules (enforced at API/permission layer, not just UI):

1. HR/admin uploads documents at enrollment (or they arrive via ATS conversion, `source=ats_import`).
2. **An employee has NO direct create/update path to `StaffDocument`.** The `documents.write` permission is not granted to the Employee role.
3. The **only** way an employee supplies/changes a document is by responding to an **HR-initiated `DocumentUpdateRequest`**. The employee's upload lands as the request's `response_file_ref` — it does **not** overwrite the live document until HR verifies.
4. HR **verifies** (`verified`) → the response is promoted to a `StaffDocument` (new/replacement) with `verification_status=verified`; or **rejects** (`rejected`) with a reason → employee may be asked again.
5. Every state change is audited; expiry alerts (M14) watch `expires_on`.

Workflow — document update (state machine):

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API:

- HR: `POST …/employees/{id}/documents` (upload), `POST …/document-requests` (raise), `POST …/document-requests/{id}/verify`, `POST …/document-requests/{id}/reject`, `…/cancel`.
- Employee (ESS): `GET …/me/document-requests`, `POST …/me/document-requests/{id}/respond` (upload only, only when status=`open`). **No** `PATCH /me/documents`.

Permissions: `documents.manage`, `documents.verify` (HR/admin, campus scope); Employee has only `documents.request.respond` scoped to self.

Acceptance:

- An Employee calling any document write endpoint outside an open request → **403**.
- HR raises a request → employee uploads → live document unchanged until HR verifies → on verify, the new file becomes the current `StaffDocument` and the old is retained in history.

- All transitions appear in the audit log.

M5 — Recruitment / ATS ★ (now MVP)

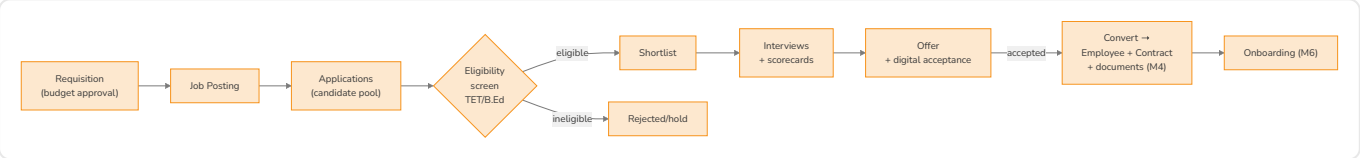
Purpose: Hire teaching/admin staff end-to-end and convert hires into the staff record without re-keying. **Actors:** HR Operator/Recruiter, Interview Panel, Approver (Principal/Group Admin), Candidate (public).

Entities: Requisition, JobPosting, Candidate, Application, Interview, Offer.

Business rules:

- Requisition needs budget-based approval (chain by amount/scope) before posting.
- **Eligibility screening:** applications are flagged against category qualification norms (TET/CTET/NET/B.Ed) at shortlist; ineligible candidates are marked, not silently dropped.
- Interview scorecards drive an outcome; offer carries CTC + `pay_group_id` + join date.
- **Offer acceptance auto-creates Employee + EmploymentContract and carries documents into M4** (as `pending` verification). No double data entry.

Workflow — hire-to-onboard pipeline:



API: `…/requisitions` (+ `/approve`), `…/job-postings`, `…/candidates`, `…/applications` (+ `/advance`, `/reject`), `…/interviews` (+ `/scorecard`), `…/offers` (+ `/send`, `/accept`), public `POST /careers/{posting}/apply`. **Permissions:** `ats.manage` (campus/group), `ats.approve` (approver), public unauthenticated apply. **Acceptance:** an approved requisition → posting → application flagged eligible → interview → offer → acceptance creates a linked Employee whose documents appear in M4 as `pending`.

M6 — Onboarding

Purpose: Convert an accepted hire (or a directly-added joiner) into an active, paid employee. **Business rules:** supports **seasonal bulk** onboarding and **academic-year fixed-term contracts with auto-expiry**; checklist tasks (documents, assets, induction); session-end **renewal/non-renewal** workflow. **Workflow:** candidate/HR data → contract → document capture (M4) → leave init (M8) → user invite → checklist. **API:** `…/onboarding-tasks`, bulk `POST …/employees:bulk`. **Permissions:** `people.manage`. **Acceptance:** a fixed-term contract auto-expires at `contract_end_date` and surfaces in the renewal queue.

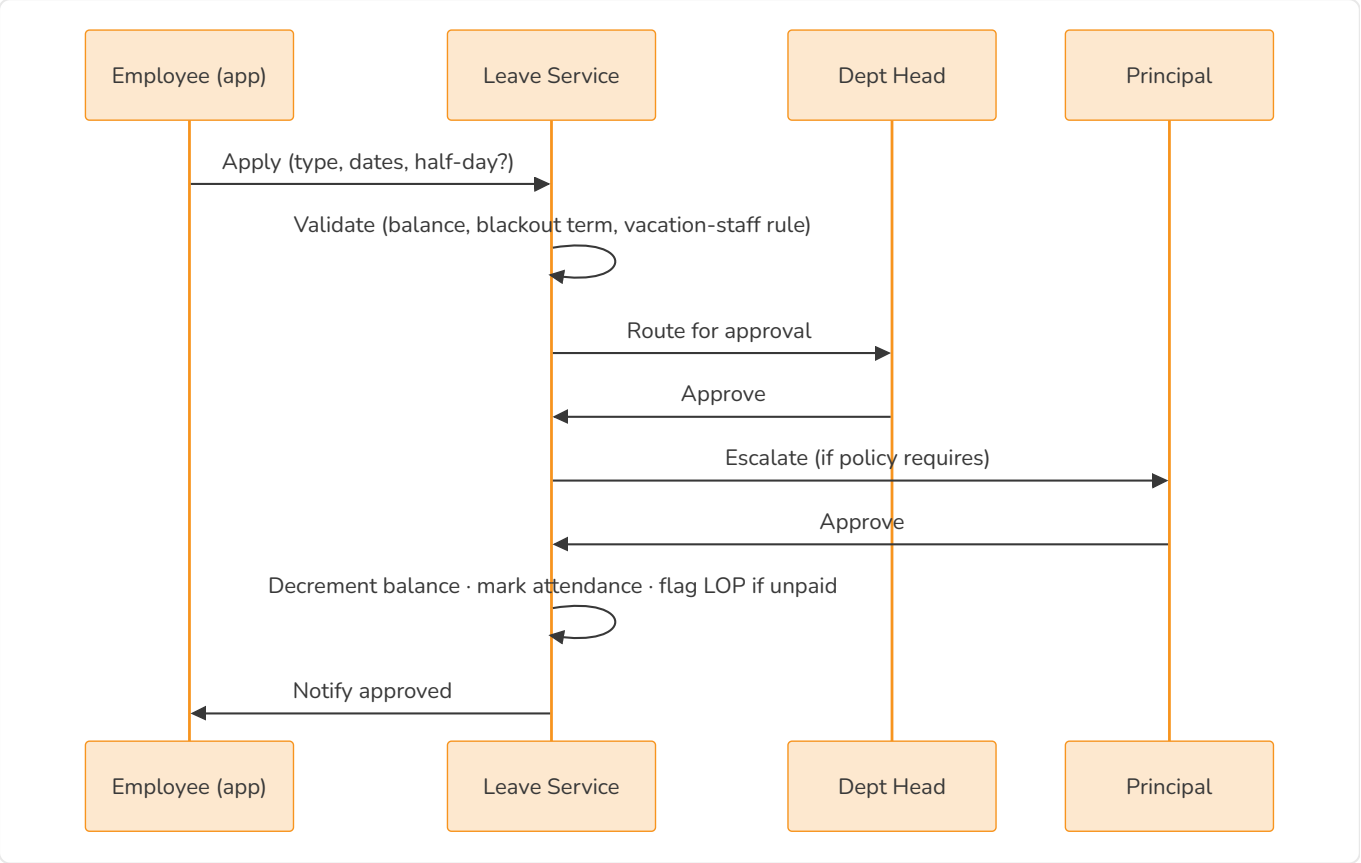
M7 — Academic Calendar

Purpose: The session/term/vacation frame that leave and (Phase 2) substitution depend on. **Entities:** AcademicYear, Term, HolidayCalendar, Holiday. **Business rules:** one `is_current` AY per scope; vacation terms feed teacher leave logic; holiday calendars are per-campus (state/festival variance). **API:** `…/academic-years|terms|holiday-calendars|holidays`. **Permissions:** `calendar.manage` (campus/group). **Acceptance:** creating AY 2026-27 with a summer vacation term makes vacation-staff leave rules apply in that window.

M8 — Leave

Purpose: Academic-calendar-driven leave with the vacation-staff distinction. **Actors:** Employee, HOD, Principal. **Entities:** LeavePolicy, LeaveType, LeaveBalance, LeaveRequest. **Business rules:** balances keyed to `academic_year_id`; `LeaveType.applies_to_vacation_staff` gates EL for teachers; exam/admission **blackout** validation against `Term`; approved paid leave → `AttendanceRecord` status; unpaid/exhausted → **LOP into payroll**; half-day supported; comp-off for holiday work.

Workflow — leave application & approval:



API: `…/leave-types`, `…/leave-balances`, `…/leave-requests` (+ `/approve`, `/reject`, `/cancel`). **Permissions:** `leave.apply` (self), `leave.approve` (HOD/Principal scope). **Acceptance:** a teacher (vacation staff) cannot accrue EL; a leave request during an exam term is blocked; an approved unpaid day reduces payroll paid-days.

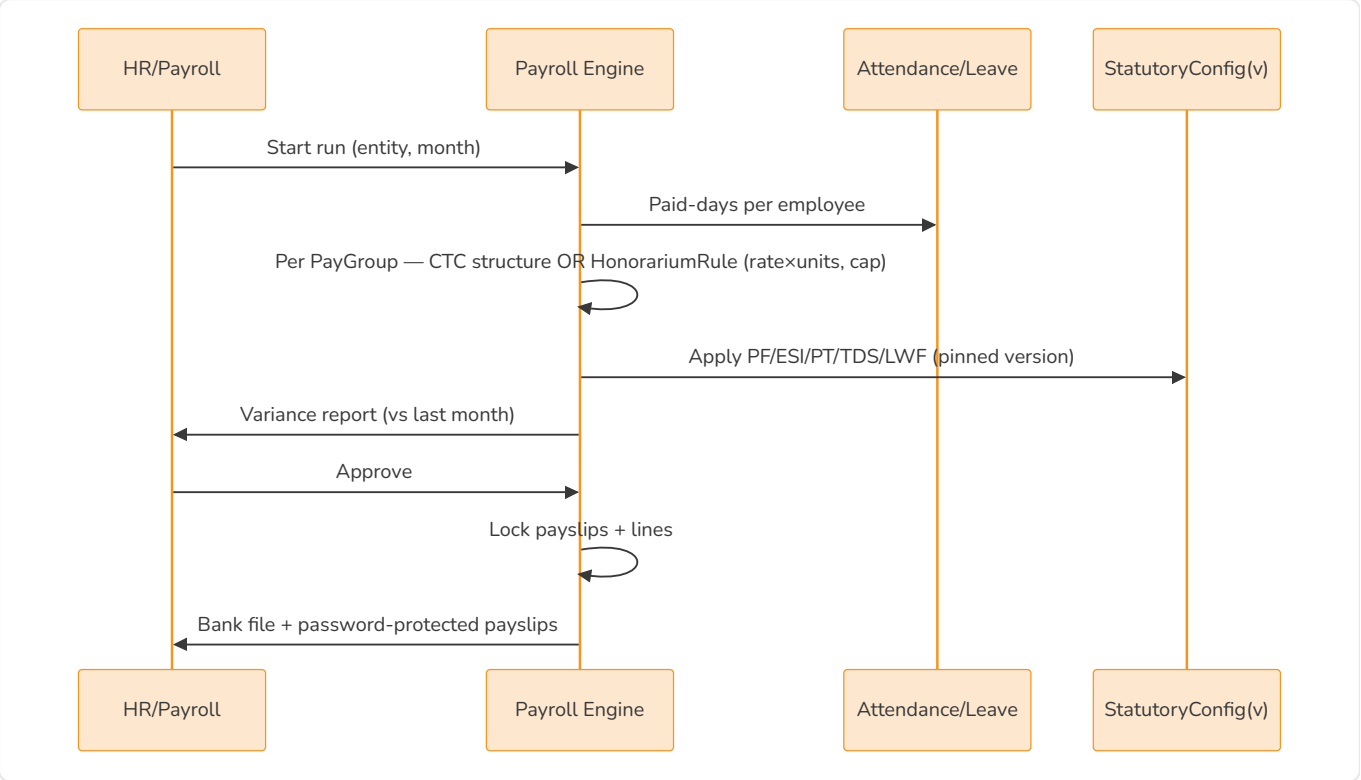
M9 — Attendance

Purpose: Multi-source capture → paid-days. **Entities:** Shift, AttendanceRecord, Regularization. **Business rules:** biometric push + mobile (geo+selfie) + web; nightly job applies shift/grace/holiday rules → status + worked_hours; exceptions resolved via Regularization; **paid-days = present + paid-leave + holidays + weekly-offs − LOP** feeds payroll. **Workflow:** device/app punch → AttendanceRecord → nightly reconciliation → exceptions → regularization. **API:** `POST …/attendance/ingest` (service auth), `…/attendance-records`, `…/regularizations`. **Permissions:** `attendance.read` (self/manager), `attendance.manage` (HR), ingest via service token. **Acceptance:** a full day's punches reconcile to `present`; a missing punch raises an exception a manager can regularize; paid-days compute correctly for a mid-month joiner.

M10 — Payroll

Purpose: Run multiple pay logics (CTC + honorarium) with statutory correctness in one entity run. **Entities:** PayGroup, SalaryStructure, PayComponent, EmployeeSalary, HonorariumRule, PayrollRun, Payslip, PayslipLine. **Business rules:** validate excluded allowances ≤ 50% of wages (Labour Code); paid-days proration; effective- dated salary; per-lecture honorarium with monthly cap; **variance review before approval**; pins `StatutoryConfig`; locked payslips; NEFT bank file.

Workflow — monthly run:



API: `…/pay-groups`, `…/salary-structures`, `…/employee-salaries`, `…/payroll-runs` (+ `/compute`, `/variance`, `/approve`, `/bank-file`). **Permissions:** `payroll.run`, `payroll.approve` (maker-checker); PII/salary gated. **Acceptance:** a run pays a CTC teacher and a per-lecture guest correctly in one pass; a back-dated revision produces arrears; excluded allowances > 50% is rejected at structure save.

M11 — Statutory Compliance

Purpose: Deep, versioned Indian statutory computation + government filings. **Entities:** StatutoryConfig, StatutoryContribution, StatutoryFiling, InvestmentDeclaration. **Business rules:** compute PF (ECR, configurable ceiling), ESI (periods rule), PT (per state), TDS §192 (old & new regime), gratuity (continuous service), bonus, LWF, min-wage; generate ECR, Form 24Q (FVU), Form 16 (Part A+B), Form 12BA (signed); reconciliation + liability provisioning; **config is versioned and pinned per run. API:** `…/statutory-configs`, `…/statutory-filings` (+ `/generate`), `…/investment-declarations`. **Permissions:** `statutory.manage`, `statutory.file`. **Acceptance:** changing the PF ceiling creates a new config version; a prior month re-opened still computes on its original version; Form 24Q validates via FVU.

M12 — Full & Final Settlement

Purpose: Correct exit settlement. **Entity:** FinalSettlement. **Business rules:** on exit — pending salary, **leave encashment (§10(10AA))**, **gratuity** (≥5 yrs, from ServiceRecord continuous service), notice recovery, loan recovery, TDS true-up; final payslip; access revoked. **API:** `…/final-settlements` (+ `/compute`, `/approve`). **Permissions:** `payroll.run` + `people.manage`. **Acceptance:** a 6-year teacher's F&F includes gratuity and leave encashment; `User.is_active` flips false on completion.

M13 — Employee Self-Service & Mobile

Purpose: Phone-first employee actions. **Business rules:** leave apply/track, attendance mark (geo/selfie) + regularize, payslip/Form 16 download, investment declaration, **respond to document-update requests (M4)** — **no unilateral document self-update**, manager approvals; native Flutter apps; offline attendance tolerance. **API:** `…/me/*` (profile read, leave, attendance, payslips, document-requests). **Permissions:** self-scoped; managers get approval permissions. **Acceptance:** an employee can respond to a doc request but receives 403 on any direct document write.

M14 — Notifications & Alerts

Purpose: Multi-channel messaging. **Business rules:** in-app/push/SMS/email; scheduled scans for **document/contract expiry** (lead times) and **statutory due dates**; approval and payslip-release events. **API:** `…/notifications`, admin templates. **Acceptance:** a licence expiring in 30 days notifies employee + HOD + HR.

M15 — Reporting & Platform

Purpose: Standard reports + audit + dashboards. **Business rules:** payroll/attendance/leave/statutory (audit-ready) reports; campus + group dashboards (headcount, cost, statutory liability); immutable audit log (read-only). **API:** `…/reports/*`, `…/audit-logs` (read). **Acceptance:** a payroll register reconciles to the sum of payslips; audit log is append-only.

M16 — Integrations

Purpose: External connectivity. **Business rules:** biometric device families (ZKTeco/eSSL/Matrix/Realtime) via push/pull ingestion; SMS/email providers; accounting export (Tally); **webhooks** (`employee.created`, `payroll_run.approved`, `offer.accepted`) for SIS/ERP; OpenAPI + generated clients. **API:** `…/webhooks`, `POST …/attendance/ingest`. **Acceptance:** a biometric push creates attendance records; a webhook fires on payroll approval.

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PART IV — ENGINEERING

7. API design & conventions

> **API layer (decision 2026-07-18): mixed GraphQL + REST.** Primary API is **strawberry-django GraphQL** (the read/query surface + most mutations, DataLoader-batched, tenancy-scoped resolvers). A **DRF REST edge** serves the cases REST fits better: device/webhook ingestion, file up/download (documents, payslips, bank files), OAuth/JWT auth endpoints, and third-party/ERP integration. **Both** feed generated clients: the GraphQL SDL → typed operations (TS via graphql-codegen, Dart via graphql codegen), and the REST edge → OpenAPI (drf-spectacular) → TS/Dart. This matches the reference stack (recoveryco).

- **Style:** GraphQL-first for app data (schema-driven, batched via DataLoaders); REST for ingestion, files, auth, and integrations. Explicit mutations/action endpoints for state transitions (`payrollRunApprove`, `POST …/payroll-runs/{id}/approve`) rather than overloaded updates.
- **Contract:** GraphQL SDL (`backend/schema.graphql`) **and** OpenAPI 3 (`drf-spectacular`) are both committed sources of truth; CI fails on drift; TypeScript + Dart clients are generated from them (never hand-written). The `check-contract-sync` hook flags stale generated clients.
- **Auth:** JWT bearer (access ~15 min, refresh rotating); every request resolves tenant + scoped roles.
- **Tenancy:** tenant is implicit from the token — **never** a request parameter a client can set.
- **Pagination:** cursor-based; **filtering/sorting** via documented query params.
- **Concurrency:** `ETag` + `If-Match` (optimistic) on mutable records; **idempotency keys** required on payroll/statutory/settlement actions so a retry never double-processes.
- **Errors:** RFC-7807 problem+json; stable machine-readable `code`; field-level validation errors.
- **Webhooks:** signed (HMAC), retried with backoff, delivered per-tenant.
- **Versioning:** URI major version (`/api/v1/…`); additive changes only within a version.

8. Security & data protection (DPDP-aligned)

Control	Implementation
Tenant isolation	ORM base manager + middleware **and** PostgreSQL Row-Level Security backstop
Encryption	TLS in transit; DB + object-store encryption at rest; password-protected payslips/Form 16
PII classification	Aadhaar, PAN, bank, salary, background-verification flagged sensitive; access-logged
Field-level access	Bank/PAN/Aadhaar/salary visibility is a distinct permission from record access
Document governance	Employees cannot write documents; only respond to HR requests (M4) — enforced server-side
Consent & retention (DPDP 2023)	Lawful-basis records, retention windows, data-access/correction requests; statutory-retention data survives exit
Child-safety	Police/background-verification records access-restricted; expiry alerts prevent lapses

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Control	Implementation
Maker–checker	Second-person approval on payroll runs and statutory filings
Auth hardening	MFA for admin/payroll roles; rate-limiting; lockout; audit of logins
On-prem	Single-tenant deployment for data-residency mandates

9. Non-functional requirements

Area	Target
Availability (SaaS)	99.5%+; payroll/statutory operations idempotent & resumable
Performance	Payroll for ~2,000 staff computes in minutes; list/search endpoints paginate and stay responsive per campus
Scalability	Scale by tenant/campus; largest single group sets the ceiling
Auditability	Every payroll run, filing, approval, document + master-data change captured immutably
Localisation	₹, Indian date/number formats, state-aware statutory, academic-year framing
Mobile offline	Attendance degrades gracefully on weak connectivity, syncs on reconnect
Maintainability	Boring stack; statutory rules in versioned data, not code
Accessibility	Web console WCAG-AA aligned for core HR flows

10. Environments, deployment & CI/CD

- **Environments:** local (Docker Compose) → staging → production (SaaS) + per-customer on-prem bundle.
- **Packaging:** containerised app + workers; the **same image** runs SaaS (multi-tenant) and on-prem (single-tenant, `deployment_mode=on_prem`).
- **CI:** lint (ruff/black, eslint) → unit + integration tests → OpenAPI diff/contract check → client-gen → migration check. Merge blocked on red.
- **CD:** staging auto-deploy on main; production gated; migrations run as a controlled step.
- **On-prem delivery:** versioned release bundle (compose/helm) + upgrade runbook; statutory-config updates ship as data so on-prem customers stay compliant without a full release.
- **Observability:** structured logs, error tracking, payroll/filing job metrics, tenant-scoped dashboards.

11. Testing strategy

- **Unit:** pure rules — statutory calculators (PF/ESI/PT/TDS/gratuity), paid-days, honorarium caps, 50%-allowance validation, leave accrual/vacation-staff logic. These carry the highest test density.
- **Integration:** API + DB with **tenant-isolation tests** (a cross-tenant read must fail against RLS), RBAC scope tests, document-governance tests (employee document-write → 403).
- **Workflow/e2e:** hire→onboard→attendance→leave→payroll→statutory→exit happy path on seed data.

- **Contract:** generated clients compile against the live OpenAPI; breaking changes fail CI.
- **Data fixtures:** a realistic multi-campus group with teaching/non-teaching/guest staff and an academic year.
- **Regression guard:** golden-file payroll outputs per statutory-config version so rule changes are intentional.

12. Build plan — milestones for ~2 engineers + AI agents

Sequenced so each stage is demoable and depends only on prior stages. Indicative sprints (2 weeks each); adjust to team velocity.

Sprint	Milestone	Delivers
0	**Foundations**	Repo, CI, Docker, Django+DRF skeleton, PostgreSQL + **RLS tenant isolation**, JWT auth, OpenAPI pipeline, client-gen
1	**Tenancy + RBAC (M1–M2)**	Org hierarchy, roles, scoped permissions, audit log, seed roles; admin console shell
2	**People + Documents (M3–M4)**	Staff master, contracts, vacation-staff flag; **document governance** (HR upload + request-gated updates + verify)
3	**ATS (M5)**	Requisition→posting→pipeline→interview→offer→**convert to employee w/ documents**
4	**Onboarding + Calendar (M6–M7)**	Seasonal/academic contracts, checklists; academic year, terms, holiday calendars
5	**Leave (M8)**	Academic-calendar leave, vacation-staff rules, approvals, blackout, LOP linkage
6	**Attendance (M9)**	Biometric ingest + mobile geo/selfie, shifts, half-day, paid-days reconciliation
7	**Payroll core (M10)**	Pay groups, structures, honorarium engine, run + variance + payslips + bank file
8	**Statutory (M11)**	PF/ESI/PT/TDS computation + versioned config; ECR, 24Q, Form 16
9	**Statutory filings + F&F (M11–M12)**	Filing artefacts, reconciliation; full & final settlement
10	**ESS + Mobile (M13)**	Flutter app: leave, attendance, payslip, **document-request responses**, approvals
11	**Notifications + Reports + Integrations (M14–M16)**	Alerts, standard reports, webhooks, accounting export
12	**Hardening**	Security review, DPDP controls, on-prem bundle, load test, golden-file payroll regression

Critical-path risks to front-load: (1) tenant isolation (Sprint 0 — everything inherits it); (2) the vacation-staff + academic-year model (Sprint 2/4 — ripples into leave & payroll); (3) versioned statutory config (Sprint 8 — must be data-versioned from first use). These are cheap now, expensive to retrofit.

Appendix — Glossary

Key terms map to the data model; see also §6 (Entity catalog).

- **Tenant** — the isolation boundary; a customer account (SaaS) or a whole on-prem install.
- **Group / Trust / Campus / Department** — the customer's internal hierarchy, *inside* a tenant.
- **Vacation staff** — employment attribute: receives vacations in lieu of earned leave (teachers). Drives leave + pay.
- **Academic year / term** — the session/vacation frame for leave (and Phase-2 substitution).
- **Pay group** — a set of staff sharing a pay logic (CTC / contract / honorarium).
- **Honorarium** — per-lecture/period/day pay for guest/substitute faculty, with monthly caps.
- **Statutory config (versioned)** — immutable snapshot of PF/ESI/PT/TDS/gratuity/bonus/LWF rules, pinned per payroll run.
- **Document update request** — the HR-initiated, verified channel through which an employee may supply/update a document.
- **LOP** — Loss of Pay; unpaid days that reduce salary.
- **F&F** — Full & Final settlement on exit (leave encashment + gratuity + recoveries + TDS true-up).

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